

ASSIGNMENT 3

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Q1. Explain in one line the difference between a class and an id selector

- A class selector (.class) is used to style multiple HTML elements that share the same class, while an ID selector (#id) is used to style one unique HTML element on a page.

Q2. Point to the <article> and the <section> on your page and say why each is the correct choice there.

- **<article>** : It is used for a self-contained piece of content that can stand on its own, such as a blog post, news story, or product review.
- **<section>** : It is used to group related content under a common topic or heading, helping organize different parts of the webpage.

Q3. Name two things semantic HTML gives you for free that a page built from <div>s does not.

- **Better accessibility:** Semantic HTML helps screen readers and assistive technologies understand the purpose and structure of the content.
- **Better SEO:** Search engines can understand the page structure and meaning more easily, which can help the page rank better.

Q4. What is the accessibility tree, and how do your tag choices change what a screen-reader user experiences?

- The accessibility tree is a simplified representation of a webpage that assistive technologies, such as screen readers, use to understand and navigate the page.
- Your HTML tag choices affect this tree—for example, using <button> tells a screen reader that an element is a button, while using a <div> may provide little or no useful information about its purpose. Semantic tags like <nav>, <main>, <article> and <heading> help screen-reader users understand the page structure and navigate it more easily.

Q5. Why is "the first rule of ARIA is: don't use ARIA" good advice? Say whether you used any, and why.

- The advice "the first rule of ARIA is: don't use ARIA" is good because native HTML elements already provide accessibility features automatically, so ARIA should only be used when necessary. I did not use ARIA because semantic HTML elements like `<button>`, `<nav>`, `<main>`, and `<article>` were enough to make the page accessible.

Q6. One line each: when is something an `<a>`, and when is it a `<button>`?

- **`<a>`**: Use it when the element takes the user to another page, section, or URL.
- **`<button>`**: Use it when the element performs an action on the current page, such as submitting, opening, or closing something.

Q7. Which two tags in your `<head>` build what a person sees in a search result, and what did you write in each?

- **`<title>`**: I wrote the page title that appears as the main clickable heading in search results.
- **`<meta name="description">`** : I wrote a short summary of the page that can appear below the title in search results.

Q8. Why is external CSS preferred over internal and inline?

- External CSS is preferred because it keeps styling separate from HTML, makes the code cleaner and easier to maintain, and allows the same CSS file to be reused across multiple pages.

Q9. Name one property that inherited from body to its children, and one that did not.

- **Inherited property**: color – it is inherited from the `<body>` to its child elements.
- **Not inherited**: margin – child elements do not automatically inherit the parent's margin

Q10. In step B23 two rules conflicted. Which won, and which of the two cascade reasons decided it?

- The class selector won over the tag selector because it has higher specificity. For example, **`.highlight`** beats `p` when both set a different value for the same property.

Q11. Describe the path from your <link> tag to a painted pixel, naming the CSSOM and the render tree.

- The <link> tag loads the CSS file, the browser parses it into the CSSOM, combines the DOM and CSSOM to create the render tree, and then uses the render tree to calculate layout and paint the final pixels on the screen.

Q12. In step B24 you misspelled a property. What did the browser do, and why is that behaviour dangerous?

- When I misspelled **color** as **colr**, the browser ignored the invalid property and continued displaying the page without applying that style. This is dangerous because CSS errors can fail silently, making mistakes difficult to notice and debug.

Q13. Name the four at-rules mentioned in the lecture and write one sentence on what each is for.

- **<header>** - It is used to contain the introductory content of a webpage, such as a logo, title, or navigation links.
- **<nav>** - It is used to contain the main navigation links that help users move around the website.
- **<article>** - It is used for a self-contained piece of content, such as a blog post, news article, or event card.
- **<section>** - It is used to group related content together under a common topic or heading.

Q14. Paste your heading outline (h1, h2, h3 only). Does the page make sense read from headings alone?

Heading Outline:

- **H1:** Campus Event Management System
 - **H2:** College Annual Fest 2026
 - **H2:** Our Events
 - **H3:** Coding Competition
 - **H3:** Dance Show
 - **H3:** Sports Events
 - **H3:** Tech Nova
 - **H3:** Robo Race
 - **H2:** Submit an Event
 - **H1:** Campus Event Management System
 - **H2:** About Our Campus Events
 - **H3:** Why Attend Campus Events?
 - **H3:** Event Highlights

- o **H2:** Event Information
 - **H3:** Registration Details
 - **H3:** Event Schedule
- o **H2:** Event Schedule
- o **H2:** Event Registration Form
- o **H2:** Useful Event Information
 - **H3:** Important Notice
 - **H3:** Technical Event
 - **H3:** Coding Instructions
 - **H3:** Keyboard Shortcut
 - **H3:** Event Reference
 - **H3:** System Message

Yes, mostly. The headings clearly describe the main topics and sections of the page, but there is one problem: the page has two <h1> headings. Ideally, it should have only one <h1> for the main page title.